

ME 150
Basics of Mechanical and Civil Engineering
Assignment#1

Instructor: Chaitanya V
Assigned Date: January 26, 2025
Due Date: January 31 2025

1. How would you categorize the below systems (closed, open, isolated). Defend your answers.



coffee



coffee



coffee



- (d) Soda
(e) Idly batter
(f) Ice cube (in deep refrigerator after frozen -assume deep refrigerator is set to be maintained at 0°C)

Fig. 1 Various thermodynamic systems

2. Describe the purpose of the below impractical thought experiment in the image below. What is its importance in engineering?

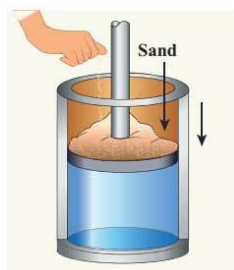


Fig. 2 A slow thought experiment

3. Consider an alcohol and a mercury thermometer that read exactly 0°C at the ice point and 100°C at the steam point. The distance between the two points is divided into 100 equal parts in both thermometers. Do you think these thermometers will give exactly the same reading at a temperature of, say, 60°C ? Explain.
4. Consider two closed systems A and B. System A contains 3000 kJ of thermal energy at 20°C , whereas system B contains 200 kJ of thermal energy at 50°C . Now the systems are brought into contact with each other. Determine the direction of any heat transfer between the two systems.
5. Categorise the below interactions (work or heat). Explain



(a) Green tea placed in a microwave (while heating)

(b) A room is heated by an iron that is left plugged in. Take the entire room, including the iron, as the system.

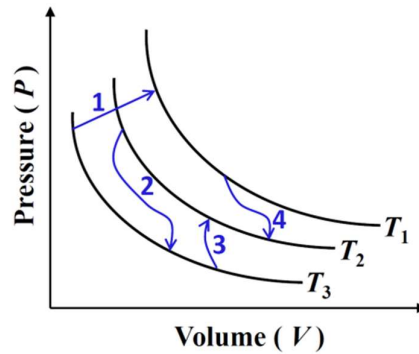
(c) A room is heated as a result of solar radiation coming in through the windows.

6. A slow chemical reaction takes place in a fluid at the constant pressure of 0.1 MPa. The fluid is surrounded by a perfect heat insulator during the reaction which begins at state 1 and ends at state 2. The insulation is then removed and 105 kJ of heat flow to the surroundings as the fluid goes to state 3. The following data are observed for the fluid at states 1, 2 and 3. For the fluid system, calculate E_2 and E_3 , if $E_1 = 0$.

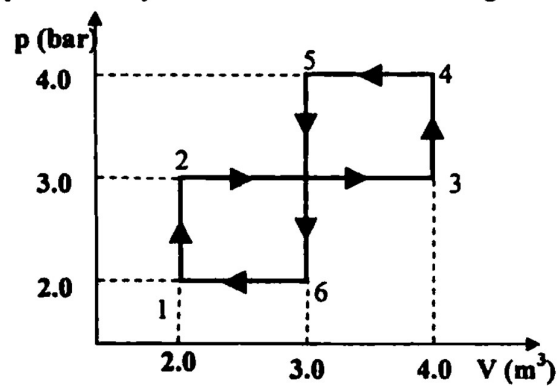
State	$v \text{ (m}^3\text{)}$	$T \text{ (}^{\circ}\text{C)}$
1	0.003	20
2	0.3	370
3	0.06	20

7. The temperature of 10 g of liquid water ($C_p = 4.2 \text{ J/g.K}$) in an insulated container is raised by 5 K by stirring. Find the amount of heat transferred to the water (in J).

8. The figure shows four different processes labelled 1, 2, 3, and 4 for the same closed system containing an ideal gas. The curves labelled T_1 , T_2 , and T_3 are isotherms. For which one of these four processes, the magnitude of internal energy change is the highest?



9. Calculate the net work output for the cycle 1-2-3-4-5-6-1 shown in below figure.



10. A gas undergoes a thermodynamic cycle consisting of the following processes:
- (i) Process 1–2: Constant pressure $p = 1.4$ bar, $V_1 = 0.028$ m³, $W_{12} = 10.5$ kJ
 - (ii) Process 2–3: Compression with $pV = \text{constant}$, $U_3 = U_2$
 - (iii) Process 3–1: Constant volume, $U_1 - U_3 = -26.4$ kJ. There are no significant changes in KE and PE.
- (a) Calculate the net work for the cycle in kJ
 - (b) Calculate the heat transfer for process 1–2
 - (c) Show that $\Sigma Q = \Sigma W$.